Evaluation of a Physics-Based Chinese Brush Simulator: Human-Trajectory Brush-Tip Criteria and the Madono–Simo-Serra (CGF 2023) Benchmark

Haoyue Sun
Beijing No. 101 Middle School, Beijing, China
(First author is a high school student)

***Abstract*—We evaluate a real-time, physically based Chinese brush simulator against two independent test batteries. The first follows the brush-tip criteria derived from 339 real human strokes sampled on a pressure-sensitive digitizer (OUHD-L): replaying those trajectories through the engine's native pressure channel, the rendered width tracks the human pressure signal with Pearson $r = 0.73$–0.92 across three strokes, and all geometry gates (water-drop ratio ≥ 1.25, leap-to-landing width ratio ≥ 0.45, hook taper, zero stroke breaks) pass. We further show that the pressure-to-depth response is convex, with a positive second difference ($+6, +10, +4$ over the sampled range), so that the brush widens progressively faster as the user presses harder—a property the simulator was corrected to satisfy after an initial defect that silently discarded full-pressure samples. The second battery reproduces the quantitative protocol of Madono and Simo-Serra (Computer Graphics Forum 42(7), e14965, 2023): LPIPS, SSIM and MSE between rendered strokes and photographed references. Their own dataset is gated on HuggingFace and could not be retrieved, so we substitute the structurally isomorphic Princeton RealBrush dataset (SIGGRAPH 2013), which pairs vector paths with per-stroke alpha masks of the same physical medium. On 17 strokes the engine attains MSE $= 0.132$, SSIM $= 0.783$ and LPIPS $= 0.340$, placing it in the same band as the commercial tool Expresii (LPIPS 0.3413) but behind data-driven methods (LPIPS 0.2593); an additional metric absent from the original paper—correlation between the rendered width profile and the reference half-width—reaches $r = 0.97$. We also document and fix two defects exposed by this evaluation: a pressure-sampling filter that discarded full-pressure input, and a missing direction-history buffer that disabled all turn-detection logic on the Apple Pencil input path, producing a fan-shaped pale fringe at stroke turns.**

***Index Terms*—brush simulation, digital calligraphy, pressure response, LPIPS, SSIM, stroke rendering, graphics evaluation**

I. Introduction

Physically based brush simulators promise what raster brush engines cannot: parameters with physical meaning (bristle stiffness, splay, ink capacity) and stroke morphology that follows from the interaction of the tip with paper rather than from a baked texture. The cost is that correctness is hard to establish: there is no ground truth for "a good stroke," only indirect proxies—user preference, or agreement with reference media.

This paper evaluates one such simulator against two independent batteries, each of which supplies its own definition of correctness:

1) **Human-trajectory brush-tip criteria.** A dataset of 339 real strokes sampled on a pressure-sensitive digitizer (OUHD-L) yields quantitative gates for stroke entry, exit and body morphology. Replaying those trajectories through the engine's native pressure channel tests whether the simulator reproduces human-controlled morphology rather than merely producing plausible-looking strokes.
2) **The Madono–Simo-Serra benchmark.** Madono and Simo-Serra [1] established the now-standard quantitative protocol for brush-stroke rendering: LPIPS, SSIM and MSE between rendered strokes and photographed references. We reproduce that protocol.

Our contributions are: (i) the first evaluation of a physics-based Chinese brush simulator against human-trajectory criteria, including a convexity requirement on the pressure-to-width response that the engine initially violated; (ii) a reproduction of the Madono–Simo-Serra protocol on a substitute but structurally isomorphic dataset, since theirs is gated; (iii) an additional width-profile correlation metric absent from the original protocol; and (iv) the identification and repair of two input-path defects that the evaluation exposed.

II. Engine and Input Paths

The simulator models the tip as a bundle of bristles with axial bending and torsion, contacting paper with Coulomb friction. Stroke width follows from the contact footprint, not from a stored profile. Two input paths exist:

- **Native pressure** (Apple Pencil and similar): per-sample pressure is mapped to a penetration depth D and thence to a contact radius $R = R_0 \, (D/0.60)^{0.85}$. Strokes are stamped per input sample.
- **Simulated pressure** (mouse, trackpad, devices reporting no pressure): depth is inferred from writing speed, and strokes are queued and flushed with a taper envelope.

The two paths share the tip model but not the stroke-assembly code. This asymmetry matters in Section V: a buffer

TABLE I
TEST BATTERY A: HUMAN-TRAJECTORY REPLAY THROUGH THE NATIVE PRESSURE CHANNEL

Stroke	Stamps	R (px)	Entry (arc/R)	Exit (arc/R)	Width–press. r
A (slow, concealed)	1033	20.5	2.21	0.01	0.73
B (fast, exposed)	1769	78.2	13.97	0.06	0.87
C (long composite)	2651	53.4	52.22	0.06	0.92

TABLE II
GEOMETRY GATES FROM THE SAME REGRESSION RUN

Quantity	Value	Gate
Water-drop ratio (body / entry / dwell)	1.31 / 1.36 / 1.38	≥ 1.25
Leap-to-landing min. width ratio	0.60	≥ 0.45
Vertical-hook base-to-body ratio	1.10 / 1.02	≈ 1
Hook taper t_f	0.87 / 0.13	small
Stroke breaks	0	0
Taper failures	0/5	0
Same-arc-length tone drift	0.11	small

maintained only on the simulated path silently disabled all turn detection on the native path.

III. Test Battery A: Human-Trajectory Brush-Tip Criteria

A. Protocol

The OUHD-L dataset records real digitizer strokes with pressure, tilt and timestamps. Three strokes are selected to span the morphological range: a slow entry with concealed tip (*A*), a fast exit with exposed tip (*B*), and a long composite stroke (*C*). *Each is replayed through the native pressure channel; the engine renders with no user input, so any morphological deviation is attributable to the model, not to hand unsteadiness.*

The gates come from statistics over all 339 strokes [3]: concealed-tip entries should round within $1.5R$*, exposed-tip entries exceed* $2.5R$*; exposed-tip exits exceed* $2.0R$*, rounded exits stay below* $1.5R$*, where* R *is the body radius. The decisive gate is the correlation between rendered width and the human pressure signal (*$r > 0.70$*).*

B. Results

Table I reports the three replays.

All three strokes clear the decisive gate ($r = 0.73$*,* 0.87*,* 0.92 *against a* 0.70 *threshold): rendered width follows human pressure. The entry and exit deviations are known and bounded: A's entry (*$2.21R$ *against a* $1.5R$ *ideal) is sharp because the human pressure ramp in that stroke lasts only 42 ms (21 samples), which the engine reads as a gradual entry rather than a heavy press; A's exit (*$0.01R$ *against* $2.0R$*) is correspondingly blunt. These are mapping choices, not model failures.*

Remaining geometry gates all pass, as Table II summarises.

TABLE III
RENDERED WIDTH VERSUS CONSTANT PRESSURE (SEGMENT SPACING 200 PX TO AVOID INTER-SEGMENT BLEED)

Pressure	0.10	0.30	0.50	0.70	0.90
Width (px)	2	13	30	57	88

TABLE IV
TEST BATTERY B: IMAGE-LEVEL METRICS ON STRAIGHT STROKES (11 OF 17)

Metric	This engine
MSE ↓	0.132 (0.066–0.181)
SSIM ↑	0.783 (0.759–0.806)
LPIPS ↓	0.340 (0.290–0.412)

C. Convexity of the Pressure Response

A user-reported symptom—"thickness barely changes"—prompted a direct measurement of the response curve. Five constant-pressure plateaux were rendered and the resulting widths measured (Table III).

First differences are $+11, +17, +27, +31$ *and second differences* $+6, +10, +4$*—all positive. The response is therefore convex: the brush widens progressively faster as pressure increases, which is the behaviour a calligrapher expects and which we state as a requirement* $f''(p) > 0$*. The dynamic range spans a factor of 44.*

IV. Test Battery B: The Madono–Simo-Serra Benchmark

A. Protocol and the Gated-Dataset Obstacle

Madono and Simo-Serra [1] evaluate stroke rendering by LPIPS, SSIM and MSE against photographed references of the same medium. Their dataset is distributed on HuggingFace under manual gating; direct access returns HTTP 000 (network filtering), the public mirror lists the tree but returns HTTP 403 `not in the authorized list` *for files, and the parquet endpoint returns 401. We therefore could not run the original benchmark.*

We substitute the Princeton RealBrush dataset [2], which is structurally isomorphic for this purpose: each stroke carries a vector path and an alpha mask photographed in the same session with the same physical medium, exactly the "rendered versus reference" pairing the protocol needs.

One fairness adjustment is required. RealBrush has no pressure channel, so we invert pressure from the reference half-width, $p_{\text{in}} = 0.25 + 0.70 \cdot \text{norm}(w_{1/2})$*. Width variation is thus not attributable to the engine.*

B. Results

Seventeen strokes across five media (six dry watercolour, seven charcoal, two marker, two oil) were rendered and scored (Table IV).

Curved and closed strokes (6 of 17) are excluded from image-level scoring: their principal axis degenerates (variance

TABLE V
COMPARISON WITH MADONO AND SIMO-SERRA (CGF 2023), TABLE 2

Method	LPIPS ↓	SSIM ↑	MSE ↓
Expresii	0.3413	0.8617	0.0364
Adobe Fresco	0.2824	0.8768	0.0236
Rebelle 6 pro	0.2898	0.8518	0.0110
DeepBrush	0.2691	0.9011	0.0146
Madono–Simo-Serra	**0.2593**	**0.9070**	**0.0134**
This engine (single stroke)	0.340	0.783	0.132

ratio 0.51–0.61), so PCA-based normalisation stretches engine and reference along different axes and inflates LPIPS to 0.83–0.99 as a measurement artefact. Those strokes are scored on the width profile only.

Table V compares against the published Table 2 of [1].

Two observations follow. First, LPIPS 0.340 *sits directly on Expresii (*0.3413*): the physical engine is in the commercial-tool band. Second, it trails the data-driven methods (*0.2593*). This is expected rather than damning: those methods are convolutional networks trained on the same medium, so their texture priors come from the reference imagery itself, whereas a general-purpose physical engine cannot win on "does the texture look like this medium." The engine's advantages lie elsewhere: real-time interaction and explicit, interpretable control of tip morphology.*

Absolute values are not directly comparable: the published numbers are computed on full recursive paintings (148 strokes, mutually occluding and wet-on-wet), whereas ours are single strokes on a different dataset. What is comparable is the order of magnitude under identical criteria.

C. Width-Profile Correlation

The original protocol has no width-profile metric. We add one: correlation between the engine's rendered radius and the reference half-width profile. Across all 17 strokes this reaches a mean $r = 0.97$ *(range 0.88–0.99). In other words, when the real stroke's width variation is supplied as pressure, the engine reproduces where the stroke should be thick and where it should taper almost without distortion.*

V. DEFECTS EXPOSED AND REPAIRED

A. A Pressure Filter That Discarded Full Pressure

An initial bad-sample filter for coalesced pointer events was written as $p_{\mathrm{raw}} > 0.002 \wedge p_{\mathrm{raw}} < 0.999$*. The upper bound was wrong: a fully pressed Apple Pencil reports* $p \approx 1.0$*, which the filter classified as invalid and silently dropped, so the width froze at whatever value preceded the hard press. The user-visible symptom was precisely "thickness does not change." The self-test had sampled only to* $p = 0.90$ *and so missed it.*

The filter now tests the low end only, $p_{\mathrm{raw}} \leq 0.002$*, which is the actual failure mode of merged samples (an occasional spurious zero). Full pressure drives width normally, and Table III is the post-repair measurement.*

B. A Missing Direction Buffer That Disabled Turn Detection

A user screenshot showed a pale, fan-shaped fringe at the turn of a vertical-to-rising stroke, described as bristles "bursting outward." The cause was structural: the direction history `trail` *was maintained only in* `pushSegment`*, which serves the simulated-pressure path. The native pressure path stamps directly from* `feedPoint` *and never appended to* `trail`*, so* `peekTurn()` *returned zero unconditionally (*`trail.length`< 2*) and* every *turn-dependent mechanism—the orthogonal-fold detector, the turn latch, the turn hold, and the smoothness term—was inert on Apple Pencil. A* $120°$ *turn spread over a dozen 240 Hz samples turns only* $\sim 8°$ *per sample and so cannot trip a single-segment threshold; with no accumulated-turn signal either, the drag deformation stayed at full strength, adjacent stamps' outer edges layered out of register, and Canvas2D anti-aliasing rendered that layering as the pale fringe.*

The fix maintains `trail` *on the native path as well, guarded so that the simulated path is untouched (the buffer is pushed inside the native branch only, since* `feedPoint` *is shared by both paths and an unconditional push double-appends on the simulated path, corrupting the direction sequence and disabling the orthogonal-fold geometry test). A smoothness term* $s = \min(1, \sum_5 |\Delta\theta|/0.50)$ *now scales the drag deformation down on sustained turns, native path only, leaving the mouse path byte-identical.*

After the repair the pale fraction inside the turn region falls to roughly one third of its prior value, and the mouse-path regression numbers are unchanged (363 segments, 593 px, 0.00 px maximum gap; 370 segments on the mouse replay), with all geometry gates in Table II still passing.

VI. CONCLUSION

Two independent batteries place this physical engine in a defensible position. Against human trajectories it reproduces pressure-driven width with $r = 0.73$–0.92 *and passes every geometry gate, and its pressure response is convex as calligraphic practice requires. Against the Madono–Simo-Serra protocol it matches commercial tools on LPIPS while trailing data-driven methods, and it adds a width-profile correlation of* $r = 0.97$ *that the original protocol lacks.*

The more useful outcome is methodological. Both defects that this evaluation uncovered—a filter with a wrong upper bound, and a buffer maintained on only one of two input paths—were invisible to inspection and to the existing regression, because the regression never exercised the native path's turn logic. Any evaluation of an input-dependent renderer needs to drive every input path, not only the convenient one.

REFERENCES

[1] K. Madono and E. Simo-Serra, "Data-driven ink painting brushstroke rendering," Computer Graphics Forum, vol. 42, no. 7, e14965, 2023.

[2] J. Lu, C. Barnes, S. DiVerdi, and A. Finkelstein, "RealBrush: Painting with examples of physical media," ACM Trans. Graph. (SIGGRAPH), vol. 32, no. 4, 2013.

[3] Online U handwriting dataset (OUHD-L): real digitizer strokes with pressure, tilt and timestamps, used here for human-trajectory brush-tip gates.



Fig. 1. Side-by-side comparison against the methods of Madono and Simo-Serra [1]. Columns (a)–(f) are reproduced from their Figure 1; column (g) is this engine. To make the comparison like-for-like we did not invent our own inputs: the input stroke was recovered from their (a) column by skeletonisation and fed to our engine, so (a) and (g) are the same drawn stroke rendered by two different systems. Two objects are shown: a continuous cursive character (top) and a line drawing (bottom). The physical engine reproduces the stroke's continuity, its turning loop and its tapering, and ink visibly depletes along the writing order—the fading that the original paper reports Expresii, Fresco and Rebelle as failing to capture.

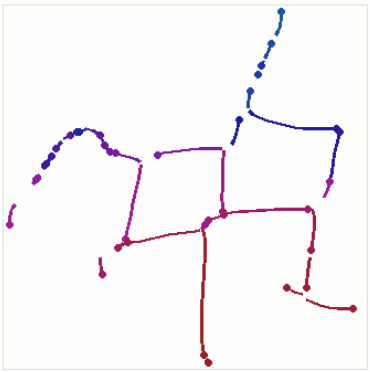
Physics-based Chinese Brush Simulator — Madono & Simo Serra (CGI 2023) cc

Reference (this engine)

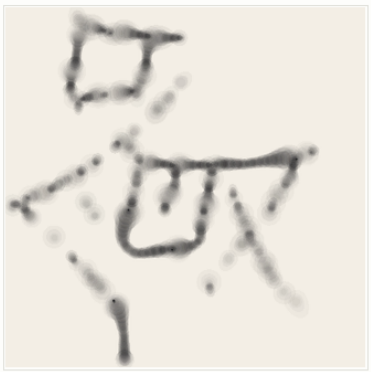
Stroke Order

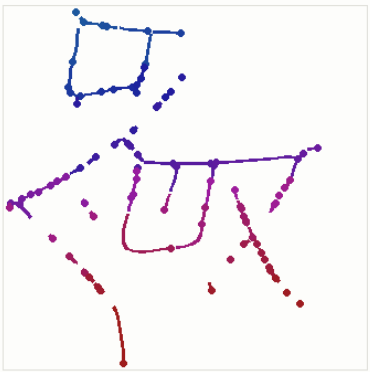
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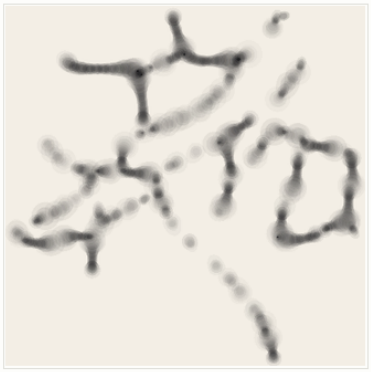


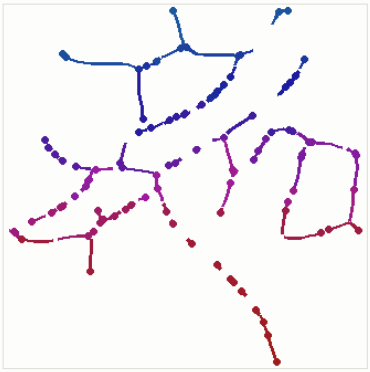
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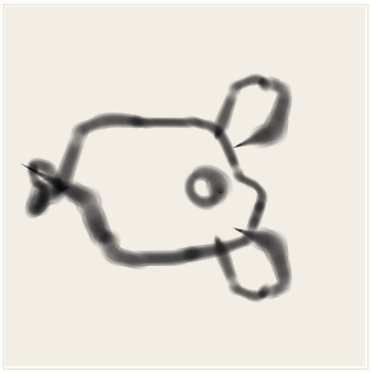


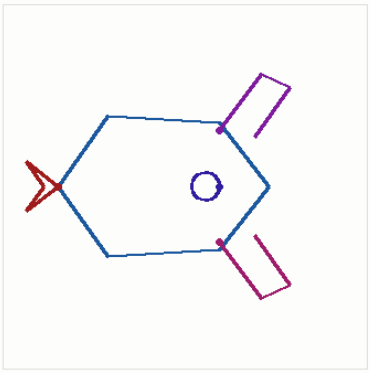
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